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A review of Bayesian group selection approaches for linear regression models

Wei-Ting Lai | **Ray-Bing Chen**

Department of Statistics, National Cheng Kung University, Tainan, Taiwan

Correspondence
Ray-Bing Chen, Department of Statistics, National Cheng Kung University, Tainan, Taiwan.
Email: rbchen@mail.ncku.edu.tw

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Abstract

Grouping selection arises naturally in many statistical modeling problems. Several group selection methods have been proposed in the last two decades. In this paper, we review the Bayesian group selection approaches for linear regression models. We start from the Bayesian indicator approach and then move to the Bayesian group LASSO methods. In addition, we also consider the Bayesian methods for the sparse group selection that can be treated as an extension of the group selection. Finally, we mention some extensions of Bayesian group selection for the generalized linear models and the multiple response models.

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KEYWORDS

Bayesian indicator selection, Bayesian LASSO, sparse group

1 | INTRODUCTION

One of the core problems of modern statistics is to identify the proper variables in the statistical model. This problem has become increasingly more important in modern applications where databases with hundreds or thousands of variables have become available, leading to a variety of methods for searching significant variables. This variable selection problem is also important in Bayesian statistics, which attempts to estimate the marginal posterior probability of variables with appropriate prior assumptions. In fact, the Bayesian variable selection approaches have been successfully applied to model selection for supersaturated designs (Beattie, Fong, & Lin, 2002; Chen, Weng, & Chu, 2013), signal processing (Févotte & Godsill, 2006 and Wolfe & Howarth, 2004), gene selection (Lee, Sha, Dougherty, Vannucci, & Mallick, 2003), and so on.

In the past two decades, many variable selection methods have been proposed. Basically, these methods can be classified into two types. One is the penalized least squares method, and the other is the Bayesian variable selection approach. The well-known method of penalized least squares is the LASSO approach, proposed by Tibshiranit (1996). This method uses the ℓ_1 penalty for the regression coefficients, and then the corresponding coefficients can be solved via an efficient convex optimization approach. Consider the Bayesian approach for the variable selection. The most commonly used approach is the stochastic search variable selection (SSVS), proposed by George and McCulloch (1993). In SSVS, an independent mixture normal distribution is chosen as the prior for each coefficient. Both normal

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distributions are centered at 0, but one has a very small variance and the other has a much larger variance. The normal distribution with a very small variance is used to model the coefficients of the variables that are not active, and the normal distribution with a very large variance is used for the coefficient of the active variable. For such a mixture of normal prior distributions, one can augment an indicator for each variable to denote whether this variable is active or not. Thus, unlike the LASSO method that targets the coefficient estimation, in SSVS, the posterior samples of the indicators are generated via a Markov chain Monte Carlo (MCMC) approach, and the best subset of the variables with the highest posterior probability can be determined as the model with active variables. Later, Park and Casella (2008) proposed the Bayesian version of the LASSO method, called Bayesian LASSO, by treating the penalty term as the conditional Laplacian prior for coefficients. Then, the posterior samples of the coefficients can be generated via an MCMC method, and the sample medians or sample means of the coefficients are reported for the posterior inference.

More recently, the works on variable selection have been explored in the "structure" selection problems. At this time, the group structure is the most indicated. That is, the variables are divided into different disjoint groups due to the pre-knowledge, and then based on the group sparsity, only a few groups are selected as active. Grouping structures can arise for many reasons, and give rise to quite different modeling goals. Grouping can also be introduced into a model in the hopes of taking advantage of prior knowledge that is scientifically meaningful. A common example is the representation of multilevel categorical covariates in a regression model by a group of indicator variables (Yuan & Lin, 2006). Consider a nonparametric additive model. The unknown component is represented by a set of B-spline basis functions, and these basis functions are treated as groups (Huang, Horowitz, & Wei, 2010). In addition, Huang, Breheny, and Ma (2012) mentioned some other applications for the group selection. For example, in genomic data analysis, the group structure is commonly used because the genes in the same pathway or a functional group are treated as a group, and the researchers are interested in selecting the genes and the corresponding pathways that affect a clinical phenotype. To deal with the group selection, the most famous method is the group LASSO, proposed by Yuan and Lin (2006), which extends the standard LASSO for group selection by replacing the penalty ℓ_1 with the ℓ_1/ℓ_2 penalty. With the Bayesian framework, the Bayesian variable selection methods, such as SSVS, can be naturally extended to group selection. Basically, the indicators are currently used to denote which "groups" are active. Farcomeni (2010) proposed a Bayesian constrained selection approach including the concept of group sparsity. Raman, Fuchs, Wild, Dahl, and Roth (2009) provided a Bayesian group LASSO by extending by the Bayesian LASSO approach.

In addition to the group selection problems in the linear regression models, some extensions are also considered in this paper. For example, we consider the group selection problems in the generalized linear models and the applications of the group selection in the multiple response models. We also mention the sparse group selection approach, an extension of the group selection, and in the sparse group selection, we do not only select the important groups but also identify the active variables within these important groups.

This paper is divided into several parts. We start from the introduction of the indicator-based Bayesian variable selection methods in Section 2. Then, we move to the Bayesian group selection approaches and try to connect the link between variable selection and group selection. In Section 3, we target the LASSO-type method, and the corresponding Bayesian LASSO methods for the group selection are introduced. Then, in Section 4, the simulation studies are used to illustrate the performances of these Bayesian group selection approaches based on the corresponding functions in different R packages. In addition to linear regression models, the Bayesian group selection methods for the generalized linear regression models and the multiple response models are discussed in Section 5. Finally, in Section 6, a short conclusion is given to summarize the group selection approaches.

2 | BAYESIAN INDICATOR SELECTION APPROACH

In this section, we start from the Bayesian variable selection approach for linear regression models, and for the given regression model, we need to specify hierarchical prior distributions over the regression coefficients that incorporate prior beliefs about the structure of the variables. As mentioned before, we first consider the simple linear regression model

$$\mathrm{Y} = \mathrm{X}\beta + \varepsilon,$$

where $\boldsymbol{Y}$ is the $n \times 1$ response vector, $\mathrm{X} = \left[\mathrm{X}_1, \mathrm{X}_2, ..., \mathrm{X}_p\right]$ is the $n \times p$ model matrix such that X_j is used to denote the jth column of variables, and ε is a noise vector with $N(\mathbf{0}, \sigma^2 I_n)$, where $\mathbf{0}$ is an $n \times 1$ zero vector, and I_n is the $n \times n$ identity matrix. In addition, β is the unknown $p \times 1$ coefficient vector.

Extending the linear regression model with group structure, the variables' X_js are divided into g nonoverlapping groups. Thus, the corresponding statistical model can be represented as

$$\mathrm{Y} = \sum_{k=1}^{g} \mathrm{X}_k \beta_k + \varepsilon, \tag{1}$$

where $\mathrm{X}_k = \left(\mathrm{X}_{k1}, ..., \mathrm{X}_{kp_k}\right)$ is the $n \times p_k$ submatrix of $\boldsymbol{X}$; $\beta_k = \left(\beta_{k1}, ..., \beta_{kp_k}\right)'$ is a vector of unknown coefficients in the kth group, and $\sum p_k = p$. Here, the goal of the group selection is to identify the active X_k, that is, the corresponding $\beta_{kj} \neq 0$ for all $j = 1, ..., p_k$.

In the following, we first introduce the SSVS approach, which is a Bayesian variable selection method, by introducing indicators to denote which variables are active and then mention how to extend SSVS for group selection problems. In addition, another indicator-type method, the Bayesian constrained selection approach, is introduced, and in this approach, the group structure is treated as one of three constraints. Finally, the sparse group selection problem is considered, and it can be treated as a generalization of the group selection because we not only select the groups but also identify the important variables within the selected groups.

2.1 | Stochastic search variable selection

George and McCulloch (1993) proposed a widely used Bayesian approach to identify the promising subset of variables in the regression models. This approach, called SSVS, employs a hierarchical Bayes mixture prior that uses the data to assign larger posterior probabilities to the more probable variables. The crucial feature for performing SSVS is the introduction of a binary latent variable (indicator) γ_j determining where the jth variable is included or excluded in the model. Thus, the prior for the jth coefficient is dependent on γ_j and is represented as

$$\beta_j \,|\, \gamma_j \;\sim\; \left(1-\gamma_j\right) N\left(0, \tau_j^2\right) + \gamma_j N\left(0, c_j^2 \tau_j^2\right).$$

Here, the prior parameters τ_j^2 and c_j^2 are chosen such that τ_j^2 is small and $c_j^2\tau_j^2$ is large. Hence, if $\gamma_j = 0$, the jth variable is not active; i.e., β_j is small and should be close to zero; that is, the prior density function of β_j tends to be close to a delta function due to the small value of τ_j^2. If $\gamma_j = 1$, then the jth variable is active; i.e., the value of β_j is large and far from zero. In addition to β_j, we also need to specify the prior distributions of γ_j and σ^2. For γ_j, its prior distribution is a Bernoulli distribution with $p(\gamma_j = 1) = p_j$ and $0 < p_j < 1$. To simplify the procedure, it is usually to set $p_j = \rho$ for all j. The inverse gamma (IG) prior is chosen for σ^2 and is the conjugate prior, that is,

$$\sigma^2 \;\sim\; IG\left(\frac{\nu}{2}, \frac{\nu\lambda}{2}\right).$$

Finally, the prior distributions of (γ_j, β_j) are assumed to be independent for $j = 1, ..., p$, and they are independent of the prior distribution of the residual variance, σ^2.

Here, we mention how to choose the proper prior parameters. For the prior probability, ρ, when we do not have prior knowledge about the data, we would set ρ as 0.5. For the parameters in the IG prior, we usually set $\nu = 0$ and λ to be any positive number, as suggested in George and McCulloch (1993). The parameters τ_j and c_j are sensitive to the selection results. As mentioned before, we should choose τ_j and c_j such that τ_j is small and $c_j\tau_j$ is large. George and McCulloch (1993) suggested a semiautomatic approach to choose the values of τ_j and c_j. The proposed approach is dependent on the interaction point and the relative height at 0 of two marginal normal densities of $N\left(0, \sigma_{\beta_j}^2 + \tau_j^2\right)$ and $N\left(0, \sigma_{\beta_j}^2 + c_j^2\tau_j^2\right)$, where $\sigma_{\beta_j}^2$ is the variance of the least squares estimation of β_j. Since both measurements are the functions of σ_{β_j}/τ_j and c_j, we can specify several candidates $\left(\sigma_{\beta_j}/\tau_j, c_j\right)$ and then identify the proper one based on these two measurements. According to George and McCulloch (1993), we can set $\left(\sigma_{\beta_j}/\tau_j, c_j\right) = (1,5)$, (1, 10), (10, 100) and (10, 500). Finally, in addition to this semiautomatic approach, we can simply use the cross-validation approach to determine the proper (τ_j, c_j).

Let $\gamma = \left(\gamma_1, ..., \gamma_p\right)$. Based on these prior assumptions, the closed form of the posterior distribution for $(\gamma, \beta, \sigma^2)$ cannot be obtained. Thus, for the posterior inference, an MCMC algorithm is used to generate the posterior samples of β, γ, and σ^2. The SSVS implements the Gibbs sampler, where each iteration samples from the conditional distributions, $[\beta|\gamma, \mathrm{Y}, \sigma^2]$, $[\gamma|\beta, \mathrm{Y}, \sigma^2]$, and $[\sigma^2|\gamma, \beta, \mathrm{Y}]$ as the following Gibbs sequence,

$$\beta^0, \left(\sigma^0\right)^2, \gamma^0, ..., \beta^{i'}, \left(\sigma^{i'}\right)^2, \gamma^{i'}, ...,$$

Then, the best subset of variables is selected according to the information contained in the posterior samples of γ. The details of this Gibbs sampler procedure are shown as follows.

With an initial value $\left(\beta^0, \left(\sigma^0\right)^2, \gamma^0\right)$, iterate the following steps.

1 The full conditional posterior distribution of β samples from a multivariate normal distribution

$$f\left(\beta^{i\prime}|\mathrm{Y}, \sigma^{i\prime-12}, \gamma^{i\prime-1}\right) \sim N_p\left(A_{\gamma^{i\prime-1}}\left(\sigma^{i\prime-1}\right)^{-2}\mathrm{X}'\mathrm{Y}, A_{\gamma^{i\prime-1}}\right)$$

where $A_{\gamma^{i'-1}} = \left(\left(\sigma^{i'-1}\right)^{-2}\mathrm{X}'\mathrm{X} + \mathrm{D}^{-1}_{\gamma^{i'-1}}\mathrm{R}^{-1}\mathrm{D}^{-1}_{\gamma^{i'-1}}\right)^{-1}$, $\boldsymbol{R}$ is the prior correlation matrix of $\beta \,|\, \gamma$ and $\mathrm{D}^{-1}_{\gamma^{i'-1}}$ is a diagonal matrix with element $(a_1\tau_1)^{-1}$, ..., $(a_p\tau_p)^{-1}$ with $a_j = 1$ if $\gamma_j = 0$, or $a_j = c_j$ if $\gamma_j = 1$.

2 Sample $(\sigma^{i'})^2$ from the full conditional posterior

$$f\left(\left(\sigma^{i\prime}\right)^2|\mathrm{Y}, \beta^{i\prime}, \gamma^{i\prime-1}\right) \sim IG\left(\frac{n + \nu_\gamma^{i\prime-1}}{2}, \frac{\left|\mathrm{Y} - \mathrm{X}\beta^{i\prime}\right|^2 + \nu_\gamma^{i\prime-1}\lambda_\gamma^{i\prime-1}}{2}\right),$$

where ν_γ and λ_γ are equal to ν and λ.

3 Sample $\gamma_j^{i'}$ from $P\left(\gamma_j^{i'} = 1|\beta^{i'}, \left(\sigma^{i'}\right)^2, \gamma_{-j}^{i'}\right)$ which is a Bernoulli distribution with probability

$$P\left(\gamma_j^{i\prime} = 1|\beta^{i\prime}, \left(\sigma^{i\prime}\right)^2, \gamma_{-j}^{i\prime}\right) = \frac{a}{a+b},$$

where

$$a = f\left(\beta^{i'}|\gamma_{-j}^{i}, \gamma_j^{i'} = 1\right) f\left(\left(\sigma^{i'}\right)^2|\gamma_{-j}^{i}, \gamma_j^{i'} = 1\right) f\left(\gamma_{-j}^{i'}, \gamma_j^{i'} = 1\right),$$
$$b = f\left(\beta^{i'}|\gamma_{-j}^{i}, \gamma_j^{i'} = 0\right) f\left(\left(\sigma^{i'}\right)^2|\gamma_{-j}^{i'}, \gamma_j^{i'} = 0\right) f\left(\gamma_{-j}^{i}, \gamma_j^{i'} = 0\right)$$

and γ_{-j} denotes that all terms of γ except to γ_j.

Once we obtain the posterior sample of γs, the active variables can be identified via the model posterior probability criterion or the median probability criterion (Barbieri & Berger, 2004). Finally, we do have a note about the prior assumption. In addition to the mixture normal prior for β_j, another commonly used prior is the spike-and-slab prior that is used as a point mass at 0 to replace the normal distribution, $N\left(0, \tau_j^2\right)$ (Geweke, 1996 and Smith & Kohn, 1996).

Based on this coefficient prior, the Gibbs sampler procedure can be simplified because once $\gamma_j = 0$, then we just simply set β_j as zero without drawing from $N\left(0, \tau_j^2\right)$, and for the other β_j with $\gamma_j = 1$, we can collect them as a vector $\tilde{\beta}$ and then sample from a multivariate normal distribution. In addition, Chen, Chu, Lai, and Wu (2011) have mentioned the componentwise Gibbs sampler to deal with this kind of prior assumption and in that method, they suggested pairwisely sampling (γ_j, β_j) for all j.

Consider the group selection problems for the model (1). It is easy to modify the SSVS to identify the active groups instead of the variables. The key is to replace the indicator γ_{kj} for $j = 1, 2, ..., p_k$ with η_k, which denotes whether the kth group is active or not. Once this indicator is equal to 1, then the priors of the coefficients within this group would follow the independent normal prior, $N\left(0, c_j^2\tau_j^2\right)$. In addition to the independent coefficient prior, we may set the prior of the coefficients that comes from a multivariate normal distribution with a zero mean vector and covariance matrix Σ_{β_k}. Then, the Gibbs sampler procedure can still be used to generate the posterior samples of the indicators, η_k.

2.2 | Bayesian constrained selection

Farcomeni (2010) first extended the traditional SSVS to a method for performing constrained variable selection. In addition to the group constraint, the hierarchical and the antihierarchical constraints are included. For these different constraints, three different indicators are used and are defined in the following.

$$\phi_k(j) = \begin{cases} 1 & \text{if } X_j \text{ is the member of the } k\text{th group} \\ 0 & \text{otherwise} \end{cases},$$

$$\zeta_{j'}(j) = \begin{cases} 1 & \text{if the } j'\text{th variable must be included in every model in which the } j\text{th is included} \\ 0 & \text{otherwise} \end{cases},$$

$$\xi_{j'}(j) = \begin{cases} 1 & \text{if the } j'\text{th variable must be excluded in every model in which the } j\text{th is included} \\ 0 & \text{otherwise} \end{cases}.$$

Since there are g disjoint groups, to have a meaningful specification of disjoint groups, the indicators ϕ_k satisfy $\sum_k \phi_k(j) = 1$ and $\sum_j \phi_k(j) \geq 1$. Note that $\phi_k, \zeta_{j'}$ and $\xi_{j'}$ are fixed and prespecified by the user. In addition to these indicators for the variables, a new latent variable η_k is used to indicate whether the kth group should be included in the final model or not. As mentioned before, if the group is included in the final model, that is, $\eta_k = 1$, all variables in the kth group will be included in the model. Let γ_j be the indicator to denote if the j variable is active or not. Therefore, if there is only a group constraint, the following equations must hold: $\gamma_j = \prod_{k=1}^{g} \eta_k^{\phi_k(j)}$. Since all but one $\phi_k(j) = 0$, $k = 1, 2, ..., g$, $j = 1, 2, ..., p$, γ_j will be equal to the latent indicator η_k for which $\phi_k(j) = 1$. Suppose the hierarchical constraint between a father variable j_1 in group k_1 and variable j_2 in group k_2 is considered. Then, the indicator for j_2 is defined as $\gamma_{j_2} = \eta_{k_1}\eta_{k_2}$ because the variable j_2 can only be included if the groups k_2 and k_1 are simultaneously in the model. After integrating with the group constraint, we have that $\gamma_{j_2} = \prod_{k=1}^{g} \eta_k^{\phi_k(j_2)} \prod_{j \neq j_2} \gamma_j^{\zeta_j(j_2)}$. When the antihierarchical constraint is considered, we can have a similar representation for the indicators. Hence, in general, γ_j can be represented as

$$\gamma_j = \left(\prod_{l \neq j} (1 - \gamma_l)^{\xi_l(j)} \gamma_l^{\zeta_l(j)} \right) \prod_{k=1}^{g} \eta_k^{\phi_k(j)}.$$

Note that the user should make sure of the hierarchical constraints that could not be replaced by the grouping constraint when identifying the indicators.

Following similar prior assumptions in SSVS, in Farcomeni (2010), the hierarchical priors for the unknown parameters and the latent variables are set as

$$\begin{aligned}&\sigma^2 \mid \eta \sim IG\left(\frac{\nu}{2}, \frac{\nu\lambda}{2}\right);\\ &\eta_k \sim Ber(w_k);\\ &\beta \mid \eta \sim N(0, \Gamma R \Gamma),\end{aligned}$$

where $\boldsymbol{\Gamma}$ is a diagonal matrix with the diagonal elements

$$\left\{\sqrt{\gamma_1\tau_1^2 + (1-\gamma_1)c_1^2\tau_1^2}, \ldots, \sqrt{\gamma_p\tau_p^2 + \left(1-\gamma_p\right)c_p^2\tau_p^2}\right\}$$

and $\boldsymbol{R}$ is a prior correlation matrix. The prior probability w_k can be interpreted as the prior probability that variables belonging to the kth group should be included in the final model. Due to the three different constraints and these prior assumptions, the marginal prior of γ_j is the Bernoulli distribution with the probability, $\prod_k w_k^{\phi_k(j)} \prod_{l\neq j}\prod_k w_k^{\phi_k(j)\zeta_j(l)}(1-w_k)^{1-\phi_k(j)\zeta_j(l)}$.

Here, to generate the posterior samples, the Gibbs sampler is still adopted and is straightforward to sample $\{\beta, \sigma^2, \gamma\}$ from the following conditional distributions

$$\begin{aligned}&\beta \mid \mathrm{Y}, \sigma^2, \eta \sim N\left(\left(\mathrm{X}'\mathrm{X} + D^{-1}R^{-1}D^{-1}\right)^{-1}\mathrm{X}'\mathrm{Y}, \sigma^2\left(\mathrm{X}'\mathrm{X} + D^{-1}R^{-1}D^{-1}\right)^{-1}\right),\\ &\sigma^2 \mid \mathrm{Y}, \beta, \eta \sim IG\left(\frac{n + \nu_\gamma^{i'-1}}{2}, \frac{\left|\mathrm{Y} - \mathrm{X}\beta^{i'}\right|^2 + \nu_\gamma\lambda_\gamma}{2}\right),\\ &\eta_k \mid \beta, \sigma \sim Ber\left(\frac{w_k a}{w_k a + (1-w_k)b}\right),\end{aligned}$$

where D is a diagonal matrix, $diag\left(\sqrt{\gamma_j\tau_j^2 + \left(1-\gamma_j\right)c_j^2\tau_j^2}/\sigma^2\right)$, $a = f(\beta|\eta_{-k}, \eta_k = 1)f(\sigma^2|\eta_{-k}, \eta_k = 1)$, and $b = f(\beta|\eta_{-k}, \eta_k = 0)f(\sigma^2|\eta_{-k}, \eta_k = 0)$. Finally, the indicator samples of γ_j are computed from η according to $\gamma_j = \left(\prod_{l\neq j}(1-\gamma_l)^{\xi_l(j)}\gamma_l^{\zeta_l(j)}\right)\prod_{k=1}^{g}\eta_k^{\phi_k(j)}$. After collecting these posterior samples of γ_js, the active groups and the corresponding variables can be identified via the estimations of the marginal probabilities of $\eta_k = 1$ and $\gamma_j = 1$ based on the median probability criterion proposed by Barbieri and Berger (2004).

Consider the setups of the prior parameters. It is similar to our discussion of the prior parameters in SSVS. For example, the prior probability w_k is set to be 0.5, and the correlation matrix R can be an identity matrix or $(\mathbf{X}'\mathbf{X})^{-1}$. For the two variances in the mixture normal priors, τ_j and $c_j\tau_j$, we still hope to choose τ_j to be small and $c_j\tau_j$ is large enough to make sure its coefficient estimation should be far from zero when this variable is active. Farcomeni (2010) suggested choosing c_j and τ_j such that $\Delta_j = \log\left(c_j^2\right)/\left(1/\tau_j - 1/\left(c_j\tau_j\right)\right)$ and $c_j\tau_j$ is large enough, where Δ_j is the ratio of the size of an insignificant change in $\boldsymbol{Y}$ and the size of a maximum feasible change in X_j.

In Farcomeni (2010), the theoretical consistency property for the model identified based on the median probability criterion has been studied. First, assume $(\mathbf{X}'\mathbf{X})/n \to C$, where C is positive definite, and assume the true and median models are included in the collection of possible models. Suppose $M_{med} \mid \mathrm{Y}$ is used to denote the median probability model and then under certain conditions, it has been shown that

$$\lim_{n\to\infty} P(M_{med} = M_0|\mathrm{Y}) = 1,$$

where M_0 is the notation for the true model. These conditions are related to prior probability, w_k, and the two variances of the normal prior distributions and may be useful to choose the proper parameters in these normal priors. For example, one condition is that $(1 - w_k)c_j\tau_j > w_k\tau_j$ for all j, and this condition would be equivalent to

$$0 < w_k < \min_j \frac{c_j^2}{c_j^2 + 1},$$

which can be a restriction for the choice of the prior probability, w_k.

2.3 | Bayesian sparse group selection

In the group selection problem, once a group is selected, all the variables within this group are included in the final model. However, sometimes the group importance may only be decided by a few variables within the group. That is, for an active group, not all variables are treated as important. For example, a drug-gene interaction study was illustrated in Chen, Chu, Yuan, and Wu (2016). In this example, the genes within the same pathway are treated as a group, and for an active pathway, only few genes are active. To deal with this kind of the scenario, Friedman et al. (2010) and Simon, Friedman, Hastie, and Tibshirani (2013) considered a sparse group selection problem, where in addition to the group sparsity, it is further assumed that within each selected group, only a small number of its constituent variables are active. Furthermore, Chen et al. (2016) studied the Bayesian sparse group selection (BSGS) problem. Unlike the single layer of indicators used in the variable and group selection approaches, in sparse group selection, they suggested using two nested layers of binary indicators. One is at the group level, where a binary variable is associated with each group, indicating whether this group is active or not. Another layer is at the level of individual variables, where a binary variable is associated with each variable, indicating whether this variable is to be selected. Thus, in the group level, we define the indicator vector, $\eta = \left(\eta_1, ..., \eta_g\right)'$, where η_k determines whether the kth group is selected or not, and for the individual variable level, $\gamma_k = \left(\gamma_{k1}, ..., \gamma_{kp_k}\right)'$ is added to denote the status of variables in the kth group. However, these two indicator vectors are not independent because once $\eta_k = 0$, all γ_{kj} values in the kth group are equal to 0; however, when $\eta_k = 1$, then we need to decide whether γ_{kj} is equal to 0 or 1.

Consider the prior assumptions in the BSGS. For the priors of these two types of indicators, first, the prior of the group indicator, η_k, is the Bernoulli distribution with $P(\eta_k = 0) = \theta_k$, and then the prior for the variable, γ_{kj}, is dependent on the group indicator prior and is defined as

$$\gamma_{kj} \mid \eta_k \sim (1 - \eta_k)\delta_0 + \eta_k Ber\left(1 - \rho_{kj}\right),$$

where δ_0 is a point mass at 0. If the kth group is not selected, that is, $\eta_k = 0$, then all indicators' γ_{kj}s are equal to 0. Otherwise, γ_{kj} follows a Bernoulli distribution with probability $P(\gamma_{kj} = 0) = \rho_{kj}$. Thus, the prior distribution of the coefficient β_{kj} is dependent on η_k and γ_{kj} defined as

$$\beta_{kj} \mid \eta_k, \gamma_{kj} \sim \left(1 - \eta_k\gamma_{kj}\right)\delta_0 + \eta_k\gamma_{kj} N\left(0, \tau_{kj}^2\right),$$

which is a kind of the spike-and-slab prior. For σ^2, the conjugate prior is chosen, that is, $\sigma^2 \sim IG\left(\frac{\nu}{2}, \frac{\nu\lambda}{2}\right)$. Finally, the prior distributions of $\left(\eta_k, \gamma_{k1}, \beta_{k1}, ..., \gamma_{kp_k}, \beta_{kp_k}\right)$, $k = 1, 2, ..., g$, are independent of each other, and when $\eta_k = 1$, the prior distributions of $(\gamma_{kj}, \beta_{kj})$s are independent for all j.

Based on these prior assumptions, Chen et al. (2016) introduced the groupwise Gibbs sampler to generate the posterior samples from the joint posterior distribution of indicators and coefficients, and the details of this MCMC procedure are shown as follows.

Algorithm 1

The groupwise Gibbs sampler for sparse group selection

1 Randomly (or systematically) select a group X_k with group size p_k. Compute $r_k = \mathrm{Y} - \sum_{l \neq k} \mathrm{X}_l \beta_l$
2 Sample η_k in terms of the following conditional probability.

$$p(\eta_k | \mathrm{Y}, \beta_{-k}, \sigma^2) = \frac{(1-\theta_k) Z_k}{(1-\theta_k) Z_k + \theta_k},$$

$$Z_k = \frac{P(\mathrm{Y} | \eta_k = 1, \eta_{-k}, \beta_{-k}, \sigma^2)}{P(\mathrm{Y} | \eta_k = 0, \eta_{-k}, \beta_{-k}, \sigma^2)} = \sum_{b \in B_k} \int \frac{P(\mathrm{Y}, \beta_k, \gamma_k = b | \eta_k = 1, \eta_{-k}, \beta_{-k}, \sigma^2) d\beta_k}{P(\mathrm{Y} | \eta_k = 0, \eta_{-k}, \beta_{-k}, \sigma^2)}$$

where $B_k = \left\{ (b_1, ..., b_{p_k}) : b_j = 0 \text{ or } b_j = 1,\ j = 1, ..., p_k \right\}$.

- if $\eta_k = 0$, then $\gamma_{kj} = 0$ and $\beta_{kj} = 0,\ j = 1, ..., p_k$.
- if $\eta_k = 1$, then run T iterations of the componentwise Gibbs sampler by Chen et al. (2011) in which the steps are similar to the groupwise Gibbs sampler. Treat r_k as a response vector and $X_{k1}, ..., X_{kp_k}$ as the candidate variables, and sample γ_{kj} and β_{kj} $j = 1, ..., p_k$ by using the Gibbs sampler.
- Given a current sample of β from the previous step, σ^2 is sampled from

$$\sigma^2 \sim IG\left(\frac{n+\nu}{2}, \frac{Res' Res + \nu\lambda}{2} \right)$$

where $Res = \mathrm{Y} - \sum_k \mathrm{X}_k \beta_k$.

To implement the groupwise Gibbs sampler in Algorithm 1, we can start from the null model. After a burn-in period, the Monte Carlo sample (η, γ) is used for the group selection and variable selection within groups. To remove the conflicting sample, that is, $\eta_k = 1$ and $\gamma_{kj} = 0$ for all j, let $\tilde{\eta}_k = \eta_k \times \mathbf{1}\left(\gamma_{kj} \neq \mathbf{0}\right)$ for the group inference. Here, the median probability criterion proposed by Barbieri and Berger (2004) is adopted to decide which groups and the corresponding variables are active. That is, the kth group is selected if the posterior estimate of $P(\eta_k = 1 | \mathrm{Y}) \geq \frac{1}{2}$ and the variable X_{kj} is selected if the posterior estimate of $P\left(\gamma_{kj} = 1 \,|\, \eta_k = 1, \mathrm{Y}\right) \geq \frac{1}{2}$. Similar to the theoretical results in Farcomeni (2010), Chen et al. (2016) also showed the consistency property for the median probability model in sparse group selection problems based on some conditions for the prior parameters. Lee and Chen (2015) provided the R package, BSGS, to implement the groupwise Gibbs sampler for sparse group selection.

Consider the setups of the tuning parameters. Two prior probabilities, θ_k and ρ_{kj}, are usually set as 0.5 due to no pre-knowledge. The parameters in the conjugate prior of σ^2 can be chosen as $\nu = 0$ and $\lambda = 1$, the same as those in George and McCulloch (1993). For the prior standard deviation of the active variables, due to the conditions for the consistency property, the lower bound of τ_{kj} is

$$\max\left\{ \frac{1-\theta_k}{\sqrt{2\pi}\theta_k}, \frac{1-\rho_{kj}}{\sqrt{2\pi}\rho_{kj}} \right\}.$$

Thus, when $\theta_k = \rho_{kj} = 0.5$, τ_{kj} should be larger than 0.4. To simplify the computation, we usually set $\tau_{kj} = \tau$. Chen et al. (2016) showed that the larger τ is, the more parsimonious the model that is obtained, and they suggested using the K-fold cross-validation method to determine the best value of τ. In addition to treating these prior parameters as the

tuning parameters, the full Bayesian approach can be adopted for the groupwise Gibbs sampler by setting the hyperpriors for the prior parameters. Following Chen et al. (2016), first, the Beta prior can be used for the probabilities θ_k and ρ_{kj}, and the hyperprior of the variance parameter τ_{kj}^2 can be an IG prior with certain parameters. After adding these hyperpriors, the groupwise Gibbs sampler would be modified by adding the steps for sampling the prior parameters. The details of the simulation results of the full Bayesian approach can be found in the supplementary materials of Chen et al. (2016).

3 | BAYESIAN GROUP SELECTION BASED ON THE LASSO-TYPE METHOD

The penalized least squares methods have been very popular for selecting relevant variables and estimating regression coefficients simultaneously. The least absolute shrinkage and selection operator, LASSO for short, was first formulated by Tibshiranit (1996). To identify the regression coefficients, the LASSO method puts a constraint on the sum of the absolute values of the model coefficients, and this sum has to be less than a fixed value. This method tries to shrink the small coefficients to be zeros, and thus the important variables can be identified due to these nonzero coefficients. The objective function of this LASSO method can be written as

$$\|\mathrm{Y}-\mathrm{X}\beta\|_2^2+\lambda\|\beta\|_1, \tag{2}$$

where λ is a tuning parameter and $\|\cdot\|_1$ stands for the vector l_1-norm and can be solved by an efficient convex optimization.

According to the objective function of LASSO in Equation (2), it can be realized as the sum of the negative log of the likelihood function and the negative log of the prior distribution for the regression coefficients. Since $\|\beta\|_1=\sum_j|\beta_j|$, we consider the prior on each coefficient as coming from the independent and identical Laplace distribution. Thus, the LASSO estimation can be considered as the maximum a posteriori estimate (Tibshiranit, 1996). Due to this connection, Park and Casella (2008) proposed the Bayesian LASSO method by setting the prior on coefficients as a conditional Laplacian prior, that is,

$$\pi(\beta|\sigma^2)\propto\prod_j\lambda\exp\left(-\lambda|\beta_j|/\sqrt{\sigma^2}\right)/\sqrt{\sigma^2},$$

and the prior of σ^2 is a noninformative scale-invariant prior, $1/\sigma^2$. Then, a Gibbs sampler procedure was proposed to generate the posterior samples of β by representing the Laplace distribution as a scale mixture of normal distributions because we do have the following representation,

$$\frac{\lambda}{2\sqrt{\sigma^2}}e^{-\frac{\lambda}{\sqrt{\sigma^2}}|\beta_j|}=\int_0^\infty\frac{1}{\sqrt{2\pi w}}e^{-\frac{\beta_j^2}{2w}}\frac{\lambda^2}{2\sigma^2}e^{\frac{-\lambda^2 w}{2\sigma^2}}dw$$

with $\frac{\lambda}{\sqrt{\sigma^2}}>0$. Due to the success of the Bayesian LASSO approach, the corresponding approach has been extended to the group (structure) selection. We introduce some of the methods in the following.

3.1 | Bayesian group LASSO methods

Yuan and Lin (2006) extended the LASSO method for group selection, and the group LASSO estimate is defined as the solution to

$$\hat{\beta}^{Group\ LASSO}(\lambda)=arg\ \min_\beta\left(\frac{1}{2}\left\|\mathrm{Y}-\sum_{k=1}^g\mathrm{X}_k\beta_k\right\|^2+\sum_k^g\lambda\|\beta_k\|_2\right), \tag{3}$$

where $\lambda \geq 0$ is a tuning parameter and $\|\cdot\|_2$ is used to denote the vector l_2-norm. They also proposed an iterative approach to find the optimal solution.

Basically, we still treat the objective function in Equation (3) as the sum of the negative log of the likelihood function with Gaussian noise and the negative log of the prior density function of the group coefficients. Following an idea similar to Park and Casella (2008), Raman et al. (2009) and Kyung, Gill, Ghosh, and Casella (2010) suggested a multivariate Laplacian prior for the group coefficients, that is,

$$\pi(\beta_k|\sigma^2) \propto \exp\left(-\frac{\lambda}{\sigma}\|\beta_k\|_2\right),$$

and this prior can be expressed as a hierarchical normal-Gamma prior. That is, we set normal priors for the regression coefficients, $\beta_k \mid \sigma^2, \tau_k \sim N\left(\mathbf{0}, \tau_k^2\sigma^2 \mathtt{I}_{p_k}\right)$, where $\mathtt{I}_{p_k}$ is a $p_k \times p_k$ identity matrix, and then a Gamma hyperprior, $Gamma((p_k+1)/2, \lambda^2/2)$, is chosen for the hyperparameter τ_k^2. Raman et al. (2009) have shown that based on this hierarchical normal-Gamma prior, the marginal prior of β_k is the multivariate Laplace prior. Finally, the Gibbs sampler can be adopted for generating posterior samples of β_k. For more details, please refer to Raman et al. (2009) and Kyung et al. (2010).

Due to the conditional normal prior for the coefficients, β_k, as shown in Raman et al. (2009) and Kyung et al. (2010), even though the group is inactive, the coefficients within this group may not be exactly zeros. To overcome this problem and increase the computational efficiency, a spike-and-slab prior, which is a mixture of the conditional normal prior and a point mass at zero, may be adopted. Therefore, for the inactive groups, we can simply set all coefficients as zeros. Xu and Ghosh (2015) proposed the Bayesian group LASSO model with spike-and-slab priors, and the corresponding priors are shown in the following:

$$\begin{aligned}
&\beta_k \mid \sigma^2, \tau_k^2 \overset{ind}{\sim} (1-\pi_0)N\left(0, \sigma^2\tau_k^2 \mathtt{I}_{p_k}\right) + \pi_0\delta_0(\beta_k);\\
&\tau_k^2 \overset{ind}{\sim} Gamma\left(\frac{p_k+1}{2}, \frac{\lambda^2}{2}\right), \quad k = 1, 2, ..., g;\\
&\sigma^2 \sim IG(\alpha, \lambda),
\end{aligned}$$

where $\delta_0(\beta_k)$ denoted a point mass at $\mathbf{0} \in R^{p_k}$. Consider the prior of $\left(\beta_k, \tau_k^2\right)$, and after integrating τ_k^2 out, $\beta_k \mid \sigma^2$ follows a mixture of the multivariate Laplace prior and point mass at $\mathbf{0}$, that is,

$$\beta_k \mid \sigma^2 \sim (1-\pi_0)M\text{-}Laplace\left(0, \frac{\sigma}{\lambda}\right) + \pi_0\delta_0(\beta_k), \tag{4}$$

where $M\text{-}Laplace(\mathrm{x} \mid 0, c^{-1}) \propto c^{p_k}\exp\left(-c\|x\|_2\right)$. For the prior parameter π_0, it may be set as 0.5, or we can set another beta hyperprior, $Beta(a, b)$. Xu and Ghosh (2015) suggest setting $a = b = 1$ in this beta hyperprior. For another prior parameter λ, instead of setting another gamma conjugate hyperprior, Xu and Ghosh (2015) took the empirical Bayesian approach and estimated the value of λ by the Monte Carlo EM algorithm with the tth update for λ as

$$\lambda^{(i)} = \sqrt{\frac{\sum_k^g p_k + g}{\sum_k^g E_{\lambda^{(i-1)}}\left[\tau_k^2 \mid \mathrm{Y}\right]}}$$

in which $E\left(\tau_k^2\right)$ is replaced by the average sample of τ_k^2 in the Gibbs sampler based on $\lambda^{(t-1)}$.

Xu and Ghosh (2015) proposed a Gibbs sampler procedure to generate the posterior samples from $P(\beta, \tau^2, \sigma^2, \pi_0 \mid \mathrm{X}, \mathrm{Y})$, and the details are shown in the following.

- Let $\mu_k = \mathbf{\Sigma}_k \mathrm{X}_k'(\mathrm{Y} - \mathrm{X}_{-k}\beta_{-k})$ and $\mathbf{\Sigma}_k = \left(\mathrm{X}_k'\mathrm{X}_k + \frac{1}{\tau_k^2}\mathtt{I}_{p_k}\right)^{-1}$; then, the conditional distribution of β_k is

$$\beta_k \mid rest \sim (1-\eta_k)N\left(\mu_k, \sigma^2\boldsymbol{\Sigma}_k\right) + \eta_k\delta_0(\beta_k),$$

where $\eta_k = P(\beta_k = 0|rest) = \frac{\pi_0}{\pi_0 + (1-\pi_0)\left(\tau_k^2\right)^{\frac{p}{2}}|\boldsymbol{\Sigma}_k|^{\frac{1}{2}}exp\left\{\frac{1}{2\sigma^2}\left\|\boldsymbol{\Sigma}_k^{\frac{1}{2}}X_k'(Y - X_{-k}\beta_{-k})\right\|_2^2\right\}}$.

- Let $\alpha_k^2 = \frac{1}{\tau_k^2}, k = 1, ..., g.$

$$\alpha_k^2 \mid rest \sim \begin{cases} IG\left(\frac{p_k+1}{2}, \frac{\lambda^2}{2}\right), & \text{if } \beta_k = 0; \\ Inverse\ Normal\left(\frac{\lambda\sigma}{\|\beta_k\|_2}, \lambda^2\right), & \text{if } \beta_k \neq 0. \end{cases}$$

$$\sigma^2 \mid rest \sim IG\left(\frac{n}{2} + \frac{1}{2}\sum_k^g p_k Z_k + \alpha, \frac{1}{2}\left[(Y - X\beta)'(Y - X\beta) + \beta' D_\tau^{-1}\beta\right] + \lambda\right),$$

where

$$Z_k = \begin{cases} 0, \ if \ \beta_k = 0; \\ 1, \ if \ \beta_k \neq 0, \end{cases} \text{ and } D_\tau^{-1} = diag\left\{\tau_1^2, ..., \tau_g^2\right\}.$$

$$\pi_0 \mid rest \sim Beta\left(a + \sum_k^g Z_k, b + \sum_k^g p_k - \sum_k^g Z_k\right).$$

Finally, note that due to the conditional prior of β_k in Equation (4), we can reparameterize it by adding an indicator, η_k, that is, $\beta_k = \eta_k \times b_k$, where

$$b_k \mid \sigma^2 \overset{ind}{\sim} M0Laplace\left(0, \frac{\sigma}{\lambda}\right), \quad k = 1, ..., g;$$
$$\eta_k \overset{ind}{\sim} Ber(1-\pi_0), \qquad k = 1, ..., g.$$

Thus, once $\eta_k = 1$, $\beta_k \mid \sigma^2$ comes from the multivariate Laplace prior, and if $\eta_k = 0$, then $\beta_k \mid \sigma^2$ places all of the mass at $\mathbf{0}$. Therefore, to obtain the posterior samples of β_k, we can sample η_k first. If $\eta_k = 1$, then the posterior sample of β_k is obtained from generating b_k. Otherwise, we can simply set $\beta_k = \mathbf{0}$. After collecting the posterior samples, the posterior mean/median can be used as the coefficient estimations.

Consider the theoretical properties for the Bayesian group LASSO model with spike-and-slab priors. Xu and Ghosh (2015) first showed that the marginal posterior median is a soft thresholding estimator. That is, when a block least estimator of a group is less than a certain threshold, the corresponding marginal posterior median of the regression coefficients in this group would be zeros. Then, they showed that the oracle property holds for the median thresholding estimator when the design matrix is orthogonal. Let M^{med} be the group model selected based on posterior median estimators and M^0 be the true group model. Xu and Ghosh (2015) proved that under certain conditions for the prior parameters, the median thresholding estimator has the oracle property. Thus, the model selection consistency is held, that is,

$$\lim_{n\to\infty} P\left(M^{med} = M^0\right) = 1,$$

and the asymptotic normality of the coefficient estimation is also satisfied.

3.2 | Bayesian sparse group LASSO

As shown in Simon et al. (2013), the sparse group penalty is a combination of the l_1/l_2 norm, as in the group LASSO, and the l_1 norm, as in the regular LASSO, and it targets both groupwise and within-group sparsity. Following ideas similar to Park and Casella (2008), Raman et al. (2009) and Kyung et al. (2010), we set the prior of each group coefficient as

$$\pi(\beta_k) \propto \exp\left(-\frac{\lambda_1}{2\sigma^2}\|\beta_k\|_1 - \frac{\lambda_2}{2\sigma^2}\|\beta_k\|_2\right), \quad k=1,...,g, \tag{5}$$

for the BSGS, and this prior can be expressed as a two-level hierarchical structure including independent **0** mean Gaussian priors on β_ks with prior parameters τ_k and η_k and the corresponding hyperpriors for τ_k and η_k. That is,

$$\begin{aligned} &\mathtt{Y} \mid \beta, \sigma^2 \sim N\left(\mathtt{X}\beta, \sigma^2 \mathtt{I}_n\right) \\ &\beta_k \mid \tau_k, \eta_k, \sigma^2 \sim N\left(0, \sigma^2 \mathtt{V}_k\right), \quad k=1,...,g, \end{aligned}$$

where $\mathtt{V}_k$ is the diagonal matrix with the diagonal elements $\left\{\left(\frac{1}{\tau_{k1}^2}+\frac{1}{\eta_k^2}\right)^{-1},...,\left(\frac{1}{\tau_{kp_k}^2}+\frac{1}{\eta_k^2}\right)^{-1}\right\}$, and the hyperprior for τ_k and η_k is set as

$$\pi\left(\tau_{k1}^2,...,\tau_{kp_k}^2,\eta_k^2\right) = c_k\left(\lambda_1^2,\lambda_2^2\right)\prod_{j=1}^{p_k}\left[\tau_{kj}^{\frac{-1}{2}}\left(\frac{1}{\tau_{kj}^2}+\frac{1}{\eta_k^2}\right)^{\frac{-1}{2}}\right]\left(\eta_k^2\right)^{\frac{-1}{2}} exp\left\{-\frac{\lambda_1^2}{2}\sum_{j=1}^{p_k}\tau_{k,j}^2 - \frac{\lambda_2^2}{2}\eta_k^2\right\}.$$

Then, based on the above prior structure, it can be shown that after integrating τ_k and η_k, the marginal prior distribution of $\beta_k \mid \sigma^2$ would be proportional to that in Equation (5). The prior of σ^2 is also the conjugate IG prior.

The hyperparameters, λ_1^2 and λ_2^2, are sensitive to the selection results because they express the prior belief of sparsity and the quantity of shrinkage. Instead of setting them as prespecified values, Xu and Ghosh (2015) suggested setting another hyperprior as follows,

$$p\left(\lambda_1^2,\lambda_2^2\right) \propto \left(\prod_k^g c_k\left(\lambda_1^2,\lambda_2^2\right)\right)^{-1}\left(\lambda_1^2\right)^p\left(\lambda_2^2\right)^{\frac{g}{2}}\exp\left\{-d_1\lambda_1^2 - d_2\lambda_2^2\right\},$$

where d_1 and d_2 are positive small constants, for example, $d_1 = d_2 = 10^{-1}$.

Due to the above prior assumptions, the mean/median estimations of the coefficients in the inactive groups may not be zeros exactly. Xu and Ghosh (2015) introduced BSGS with the spike-and-slab-type prior for the coefficients. First, they separately reparameterize the coefficient vectors due to the two kinds of sparsity, that is, $\beta_k = \mathtt{V}_k^{\frac{1}{2}}\mathtt{b}_k$, where $\mathtt{V}_k^{\frac{1}{2}} = diag\left\{\tau_{k1},...,\tau_{kp_k}\right\}, \tau_{kj} \geq 0, k=1,...,g; j=1,...,p_k$. Thus, $\mathtt{V}_k$ is used to control the size of the $\mathtt{b}_k$ because when $\tau_{kj}=0$, β_{kj} must be equal to zero. For $\mathtt{b}_k$, its prior is a spike-and-slab prior, which is a mixture of the multivariate normal prior and point mass at **0**,

$$\mathtt{b}_k \overset{ind}{\sim} (1-\pi_0)N_{p_k}\left(\mathbf{0}, \mathtt{I}_{p_k}\right) + \pi_0\delta_0(\mathtt{b}_k).$$

Due to the sparsity within the group, the prior of τ_{kj} is also another spike-and-slab prior,

$$\tau_{kj} \overset{ind}{\sim} (1-\pi_1)N^+\left(0,s^2\right) + \pi_1\delta_0\left(\tau_{kj}\right), \quad k=1,...,g;\ j=1,...,p_k,$$

where $N^+(0, s^2)$ is a normal density function with a mean of 0 and variance s^2 truncated from below by zero.

According to the two sparse assumptions, two indicators (latent) are added into the model, and the coefficient β_{kj} is represented as

$$\beta_{kj} = \eta_k \gamma_{kj} b_{kj},$$

where η_k is the indicator for the kth group and γ_{kj} is the indicator to denote if β_{kj} is equal to zero or not. Here, we set the priors of η_k and γ_{kj} as the independent Bernoulli priors with probabilities π_0 and π_1 separately, and b_{kj} comes from a normal prior, $N(0, s^2)$. That is,

$$\begin{aligned}
&\eta_k \sim Ber(\pi_0), \quad k = 1,2,...,g, \\
&\gamma_{gj} \sim Ber(\pi_1), \quad k = 1,2,...,g \;\; j = 1,2,...,p_k, \\
&b_{gj} \sim N\left(0, s^2\right), \quad k = 1,2,...,g \;\; j = 1,2,...,p_k.
\end{aligned}$$

For the prior for σ^2, it is set as

$$\sigma^2 \sim IG(\alpha, \lambda).$$

The positive hyperparameters are usually set as $\alpha = 0.1$ and $\lambda = 0.1$. Consider the hyperparameters π_0, π_1 and s. The hyperpriors are chosen as follows,

$$\pi_0 \sim Beta(a_1.a_2); \pi_1 \sim Beta(c_1, c_2); s^2 \sim IG(1, t).$$

Instead of prespecifying the value of t, Xu and Ghosh (2015) suggested using the Monte Carlo EM to estimate t.

Based on these prior structure assumptions, Xu and Ghosh (2015) introduced a Gibbs sampler to simulate the posterior distribution, $P(\mathtt{b}, \tau^2, \sigma^2, \pi_0, \pi_1, s^2 \mid \mathtt{X}, \mathtt{Y})$. The details of the proposed Gibbs sampler are shown in the following.

- Let $\mu_k = \mathbf{\Sigma}_k \mathtt{X}_k' \left(\mathtt{Y} - \mathtt{X}_{-k} \mathtt{V}_{-k}^{\frac{1}{2}} \mathtt{b}_{-k}^{\frac{1}{2}}\right)$ and $\mathbf{\Sigma}_k = \left(\frac{1}{\sigma^2} \mathtt{V}_k^{\frac{1}{2}} \mathtt{X}_k' \mathtt{X}_k \mathtt{V}_k^{\frac{1}{2}} + \mathtt{I}_{p_k}\right)^{-1}$. Then,

$$\mathtt{b}_k \mid rest \sim (1 - l_k) N(\mu_k, \mathbf{\Sigma}_k) + l_k \delta_0(\mathtt{b}_k),$$

where $l_k = P(\beta_k = 0 | rest) = \dfrac{\pi_0}{\pi_0 + (1-\pi_0)|\mathbf{\Sigma}|^{\frac{1}{2}} \exp\left\{\frac{1}{2\sigma^4}\left\|\mathbf{\Sigma}_k^{\frac{1}{2}} \mathtt{V}_k^{\frac{1}{2}} \mathtt{X}'_k \left(\mathtt{Y} - \mathtt{X}_{-k} \mathtt{V}_{-k}^{\frac{1}{2}} \mathtt{b}_{-k}^{\frac{1}{2}}\right)\right\|_2^2\right\}}$.

- The conditional posterior of τ_{kj} is a spike-and-slab distribution,

$$\tau_{kj} \mid rest \sim \left(1 - q_{kj}\right) N^+\left(u_{kj}, \nu_{kj}^2\right) + q_{kj} \delta_0(\tau_{kj}),$$

where $q_{kj} = P\left(\tau_{kj} = 0 | rest\right) = \dfrac{\pi_1}{\pi_1 + (1-\pi_1)(s^2)^{\frac{-1}{2}} \left(\nu_{kj}^2\right)^{\frac{-1}{2}} exp\left\{\frac{u_{kj}^2}{2\nu_{kj}^2}\right\}\left[\Phi\left(\frac{u_{kj}}{\nu_{kj}}\right)\right]}$; $u_{kj} = \frac{1}{\sigma^2} \nu_{kj}^2 \left(\mathtt{Y} - \mathtt{X}_{k,-j} \beta_{k,-j}\right)'$ and $\nu_{kj}^2 = \left(\frac{1}{s^2} + \frac{1}{\sigma^2} \mathtt{X}_{kj}' \mathtt{X}_{kj} b_{kj}^2\right)^{-1}$.

Here, $\mathtt{X}_{k,-j}$ denote the model matrix of $\mathtt{X}_k$ by removing the jth column, $\mathtt{X}_{kj}$, and $\beta_{k,-j}$ is the coefficient vector of the kth group without the jth component.

$$\sigma^2 \mid rest \sim IG\left(\frac{n}{2} + \alpha, \frac{1}{2} \|\mathtt{Y} - \mathtt{X}\beta\|_2^2 + \lambda\right)$$

- The posteriors of π_0 and π_1 conditional on everything else to be beta distributions:

$$\pi_0 \mid rest \sim Beta(\#(b_k = 0) + a_1, \#(b_k \neq 0) + a_2);$$
$$\pi_1 \mid rest \sim Beta\big(\#(\tau_{kj} = 0) + c_1, \#(\tau_{kj} \neq 0) + c_2\big).$$

- The conditional posteriors of s^2 follow the IG distribution:

$$s^2 \mid rest \sim IG\left(1 + \frac{1}{2} + \#(\tau_{kj} = 0), t + \frac{1}{2}\sum_{k,j}\tau_{kj}^2\right).$$

After the burn-in, the remaining samples are treated as the samples from the true posterior distribution, and then the posterior inference can be accomplished via the posterior mean/median of the posterior samples.

4 | SIMULATIONS

There are some R packages designed for Bayesian group selection and sparse group selection approaches. Here, two of them are used to illustrate their performances in the corresponding structure selection problems. The first package is the "MBSGS" and involves the functions for the Bayesian group LASSO approach and the Bayesian sparse group LASSO approach based on the algorithms proposed by Xu and Ghosh (2015). For the indicator selection approach, the R package, BSGS, is considered and its function, BSGS.Simple, implements the BSGS approach in Chen et al. (2016). To demonstrate the performances of these functions, based on the linear regression model, two different scenarios with the different group structures in Equation (1) are considered. The first scenario is related to the sparse group structures, and in the second scenario, the group selection problem is considered.

We start from introducing these scenarios in the following.

Scenario 1 A sparse group selection example in Chen et al. (2016) is revisited. Based on the linear model (1) assumption, there are 40 observations and 60 variables divided into six groups, G_1, G_2, ..., G_6 with different group sizes. In the first two groups, G_1 and G_2 have five variables in each group. G_3 and G_4 have 10 variables. There are 15 variables in G_5 and G_6. The variables are generated from the multivariate normal distribution with zero mean and variance 1.25, and correlation between two different variables is 0.2 if both variables are in the same group, and 0 otherwise. There are five active variables, X_3, X_{11}, X_{12}, X_{31}, and X_{32} with coefficients $\beta_3 = 3.2$, $\beta_{11} = -2$, $\beta_{12} = 1$, $\beta_{31} = 1.5$, $\beta_{32} = -1.5$, respectively. The error term ε follows a normal distribution with zero mean value and its variance is equal to 4. Thus, $G1$, $G3$ and $G5$ are active groups. After learning the model based on the 40 observations, 10 more points are generated for checking the prediction ability.

Scenario 2 The group selection problem is considered in this example, and it is a large p small n problem by setting $n = 50$ and $p = 200$. These 200 variables are divided into 40 groups, and there are 5 variables in each group. Following Chen et al. (2011), the jth variable in the group k is defined as $X_{kj} = z_k + z_{kj}$, where z_k and z_{kj} are from an independent standard normal distribution, $k = 1, 2, ..., 40$; $j = 1, 2, ..., 5$. Thus, variables within a group are correlated with pairwise correlation 0.5, while the variables in different groups are independent. The coefficient of β is set as

$$\beta = ((3, -3.5, 4, -2.8, 3.2), \mathbf{0}, \mathbf{0}, (3, -3.5, 4, -2.8, 3.2), \mathbf{0}, \mathbf{0}, ..., \mathbf{0}, \mathbf{0}, \mathbf{0}),$$

where $\mathbf{0}$ is a zero vector of length 5. Therefore, the first and the fourth groups are important groups, and the other groups are treated as inactive. The random error in the linear model is sampled from a normal distribution with mean zero and variance 4. To check the prediction performance, we would independently generate 10 more observations.

Here three R functions for the different Bayesian group selection approaches are considered and the set-ups of the corresponding tuning parameters are shown in the following:

Bayesian sparse group selection (BSGS): To implement the BSGS in Chen et al. (2016), the function, BSGS.Simple, in the R package "BSGS" is adopted and the corresponding tuning parameters are set as follows. First $\theta_k = 0.5$, $k = 1, 2, ..., g$; $\rho_{kj} = 0.5$, for all j and k; $\nu = 0$ and $\lambda = 1$ for the IG prior of σ. The variance of the normal prior τ^2_{kj} is fixed as 1.5 for the first scenario and as 12.25 for the second one. Here, we run 3,000 sweeps for sparse group variable selection, and for the componentwise Gibbs sampler for variable selection within a group, we also run 3,000 sweeps to generate the posterior samples. Then, the median probability criterion is used to identify the active groups and the corresponding variables within these active groups.
Bayesian group LASSO with spike-and-slab priors (BGLSS): Consider the Bayesian group LASSO with spike-and-slab priors in Xu and Ghosh (2015). The function, BGLSS, in the R package, "MBSGS", is used. To run this function, the hyperprior parameters for π_0 are fixed as $a = 1$ and $b = 1$, and for the parameters in the prior of σ^2, we set $\alpha = 0.1$ and $\lambda = 0.1$. In this function, we set to run 10,000 iterations, in which the first 5,000 are burn-ins and the remaining samples are treated as the posterior samples. Finally, the posterior median is used for Bayesian inference to determine the active variables.
BSGS with the spike and slap priors (BSGSSS): The function, BSGSSS, in the R package, "MBSGS", is used to implement the BSGS with the spike and slap priors (Xu & Ghosh, 2015). The corresponding hyperprior parameters, a_1, a_2, c_1 and c_2, for π_0 and π_1 are all set as 1. The parameters in the prior σ^2 are the same as the ones used in BGLSS. In addition, we also run 10,000 iterations and after discarding the first 5,000 samples, the remaining samples are treated as the posterior samples for the.

For comparison purposes, we also implement the functions for the sparse group LASSO and group LASSO in the R packages, "SGL" and "grpreg," separately.

- *Group LASSO (GL)*:To implement the group LASSO in Yuan and Lin (2006), the "grpreg" function with the grLasso penalty in the R package,"grpreg", is used. The objective function is the same as Equation (3) and the corresponding parameter λ is first tuned via a 10-fold cross-validation approach, and then we fix this λ value as 0.5 for both scenarios. In addition, we set the maximum number of iterations at 100 in this function.

Sparse group LASSO (SGL):For the sparse group LASSO proposed by Simon et al. (2013), the "SGL" function in the R package, SGL, is used. There are two tuning parameters, λ and α in the corresponding objective function as shown in Simon et al. (2013). The weighted parameter α decides the balance among the element-wise sparsity and group sparsity, and is simply fixed at 0.95 due to the suggestion by Simon et al. (2013). For another parameter λ, the degree of sparsity of the solution, it is first tuned via a 10-fold cross-validation, and we choose the λ as 0.08 for the first scenario and set it as 0.02 for the second scenario. In addition, we also set the maximum number of iterations at 100.

To show the performances of these selection methods, we consider the following three measures in the simulation studies: the true classification rate (TCR), the true positive rate (TPR), the false positive rate (FPR), and the false discovery rate (FDR) which are defined as follows:

$$\text{TCR} = \frac{\text{number of correctly selected variables}}{\text{number of variables}},$$

$$\text{TPR} = \frac{\text{number of correctly identified active variables}}{\text{number of true active variables}},$$

$$\text{FPR} = \frac{\text{number of incorrectly identified active variables}}{\text{number of true inactive variables}},$$

$$\text{FDR} = \frac{\text{number of incorrectly identified active variables}}{\text{number of identified active variables}}.$$

TABLE 1 The selection results for five methods in two scenarios

	Method	TPR	FPR	TCR	FDR	MSE
Scenario 1						
	BSGS	92%	14.96%	85.65%	46.41%	8.14
		(0.14)	(0.23)	(0.21)	(0.06)	(6.10)
	BSGSSS	86%	3.09%	96.91%	22.59%	6.99
		(0.18)	(0.03)	(0.03)	(0.20)	(3.99)
	BGLSS	67%	44.29%	56.67%	78.94%	17.74
		(0.39)	(0.32)	(0.27)	(0.22)	(10.13)
$\lambda = .08$	SGL	89%	13.15%	87.05%	59.54%	9.04
		(0.13)	(0.05)	(0.05)	(0.12)	(4.83)
$\lambda = .5$	GL	87%	51.20%	51.97%	55.92%	14.03
		(0.20)	(0.17)	(0.15)	(0.03)	(6.61)
Scenario 2						
	BSGS	100%	0.29%	99.75%	3.49%	6.14
		(0.00)	(0.01)	(0.01)	(0.09)	(4.77)
	BSGSSS	100%	0.13%	99.88%	1.95%	5.59
		(0.00)	(0.00)	(0.00)	(0.06)	(3.04)
	BGLSS	100%	0.79%	99.25%	3.62%	6.50
		(0.00)	(0.04)	(0.04)	(0.16)	(8.51)
$\lambda = .02$	SGL	94%	22.74%	78.11%	81.85%	53.49
		(0.08)	(0.03)	(0.03)	(0.03)	(46.00)
$\lambda = .5$	GL	100%	4.00%	96.20%	35.49%	9.13
		(0.00)	(0.03)	(0.03)	(0.23)	(6.58)

Note: The five methods in the simulation are Bayesian sparse group selection (BSGS) in Chen et al. (2016), BSGS wit spike and slap priors (BSGSSS) in Xu and Ghosh (2015), Bayesian group LASSO with spike and slap priors (BGLSS) in Xu and Ghosh (2015), sparse group LASSO (SGL) and group LASSO (GL). In addition, the standard deviations of the measures are shown in brackets.

Abbreviations: FDR, false discovery rate; FPR, false positive rate; MSE, mean squared error; TCR, true classification rate; TPR, true positive rate.

TCR is an overall evaluation of accuracy in the identification of the active and inactive variables. TPR is the average rate of active variables that are identified correctly and is used to measure the power of the method. FPR is the average rate of inactive variables in the final model, and it can be viewed as the type I error rate of the selection method. FDR is the proportion of incorrectly identified active in all identified active variables, and it is used to measure the probability that a selected variable is a false positive. Since we generate 10 more data points for each scenario, thus we also report the mean squared error (MSE),

$$\mathrm{MSE} = \frac{\sum_{i=1}^{10} (y_i - \hat{y}_i)^2}{10},$$

where $\hat{y}_i$ is the prediction of y_i.

Here, for each scenario, we repeat the simulation 100 times by independently regenerating the data due to the simulation model. Based on 100 replications, the average values of these measures with respect to the different methods are reported in Table 1. Scenario is designed as a sparse group selection problem. Both methods, BGLSS and GL, have

TABLE 2 The total running time (seconds) of five methods for 100 replications in two scenarios

Method	BSGS	BSGSSS	BGLSS	SGL	GL
Scenario	128,404.11	1,571.23	970.41	1.23	0.32
Scenario	13,711.69	12,423.79	6,812.58	20.89	2.67

Abbreviations: BGLSS, Bayesian group LASSO with spike and slap; BSGS, Bayesian sparse group selection; BSGSSS, Bayesian sparse group selection wit spike and slap; GL, group LASSO; SGL, sparse group LASSO.

higher FPR and FDR values because once a group is selected, all the variables are treated as important. Thus, the findings indicate serious overselection problems. For these sparse group selection approaches, BSGS, BSGSSS, and SGL have higher TPR and TCR values and lower FPR and FDR values. These three methods also have better prediction performance because of the lower MSE values. Among them, BSGSSS performs best because of lower FPR, FDR, and MSE values and in addition, BSGSSS also has the lower standard deviations for all these measures. In Scenario , all methods have similar TPR values that are close to 100%. However, these sparse group selection methods have overselection problems. For SGL, its FPR value is close to 23%, and it has the worst MSE value. BSGS and BSGSSS only show slight overselection because their FPR values are less than 0.5% and FDR values are less than 4%. Since it is a group selection problem, group selection approaches perform well. Here, BGLSS almost identifies the true model in every replication and its FDR value is also less than 4%. The group LASSO, GL, achieves 100% in TPR and a lower FPR value.

In addition to compare the selection performances among the different approaches, we also report the computational costs for all five methods in Table 2 and all the codes are run on an ASUS workstation equipped with Intel(R) Xeon(R) CPU E3-1245 v6 @ 3.70 GHz. Of course, LASSO type methods take the advantage in the computational cost. Due to the Markov chain structure in the MCMC algorithm, the Bayesian approaches would take more computational cost. In these three Bayesian approaches, BSGS takes the longest CPU time for both scenarios because BSGS involves two nested "for" loops in its code.

Overall, Bayesian approaches do have better selection results in both scenarios even though they take more CPU time. Among these three Bayesian approaches, BSGSSS is more stable in both cases. LASSO-type methods show the overselection in both scenarios. In fact, our simulation results also support the advantages of BSGSSS mentioned in Xu and Ghosh (2015) because the BSGSSS does have lower FPR and FDR values and is less sensitive to the model structure. To improve the selection results, determining how to tune the parameters for these functions should be an issue. At the least, cross-validation should serve as a possible method to perform parameter tuning even for Bayesian approaches. Finally note that we provide the R code for the simulation study and the readers can find it from the Supporting Information.

5 | EXTENSIONS FOR OTHER MODELS

In this section, we mention some extensions of Bayesian group selection with respect to the different models and problems. These extensions would cover the group selection in the generalized linear models and the multiple response models.

5.1 | Extensions for generalized linear models

5.1.1 | Binary response

First, we consider the binary response model. Chang, Chen, and Chi (2016) suggested taking the probit model, and they generalized the componentwise Gibbs sampler for the variable selection in the binary response model. Suppose Z_j is used to denote the binary response. Let Y_j be a latent variable, and $Y_j \mid \beta$ comes from a normal distribution with mean $x_j\beta$ and unit variance, where x_j is the jth observation of $\mathbf{X}$. Chen et al. (2016) modified the groupwise Gibbs sampler to deal with the sparse group selection in the binary response model. Basically, they add one step to generate Y_j from

$N_{Z_j}\left(x'_j\beta,1\right)$ by fixing $\sigma^2=1$, where N_{Zj} is the normal distribution with left truncation at 0 if $Z_j=1$, and N_{Zj} is the normal distribution with right truncation at 0 when $Z_j=0$. The same idea can be used for the group selection in the probit model.

5.1.2 | Count data

Raman et al. (2009) provided the Bayesian group LASSO method to analyze count data in the contingency table by modeling count data via a Poisson regression that involves a log-linear model with an independent term. That is, for $j=1,\ldots,n$, $y_j \mid \mu_j \sim Poi(\mu_j)$, with the Poisson mean $\mu_j=e^{\eta_j}$ and

$$\eta_j=\mathrm{x}'_j\beta+\varepsilon_j,\quad \varepsilon_j\sim N\left(0,\sigma^2\right).$$

Raman et al. (2009) add one step in the Gibbs sampler for the Bayesian group LASSO, and in this step, given the other components that group each main effect and the interaction effects, η_j is generated from the conditional density,

$$p\left(\eta_j\mid\beta,\sigma^2,\mathrm{X},\mathrm{Y}\right)\propto\exp\left(\sum_j\eta_j y_j-\exp\left(\eta_j\right)-\frac{1}{2\sigma^2}\left(\mathrm{x}\prime_j\beta-\eta_j\right)^2\right),$$

and they proposed a method based on the Laplace approximation instead of the adaptive rejection sampling.

5.1.3 | Cox proportional hazard model

Gu, Yin, and Lee (2013) proposed a Bayesian two-step LASSO procedure for biomarker selection under the Cox proportional hazards model,

$$h\left(t|x_j\right)=h_0(t)\exp\left\{\mathrm{x}'_j\beta\right\},$$

where $h_0(t)$ is the baseline hazard function and β is a vector of the unknown parameter. In the first step, they group each biomarker's main effect and the interaction effects of each biomarker with treatments, and the Bayesian group selection for the Cox proportional hazard model is adopted to generate the posterior coefficient samples. Then, a loose selection criterion is computed based on these posterior samples and is used to screen out these insignificant biomarkers. Based on the remaining biomarkers, a Bayesian adaptive LASSO is used to select the individual markers and interactions between markers and treatments in order to identify predictive markers.

5.2 | Extensions for multiple response linear models

5.2.1 | Support union recovery

Chen, Wu, and Chen (2014) considered the Bayesian structure selection problems in the multiple response model, that is,

$$\mathrm{Y}=\mathrm{X}B+W, \tag{6}$$

where $\mathrm{Y}=[\mathrm{Y}_1,\ldots,\mathrm{Y}_M]$ is an $n\times M$ response matrix of observations, $\mathrm{X}=\left[\mathrm{X}_1,\ldots,\mathrm{X}_p\right]\in\mathbb{R}^{n\times p}$ is the fixed $n\times p$ design matrix, $B=[\beta_1,\ldots,\beta_M]$ is a $p\times M$ matrix of the unknown regression coefficients, and $W=[\omega_1,\ldots,\omega_M]\in\mathbb{R}^{n\times M}$ is the corresponding

noise matrix. Here, the error term ω_m is an $n \times 1$ noise vector that follows a multivariate normal distribution with zero mean vector and covariance matrix $\sigma^2 \mathtt{I}_n$. We assume that these response vectors may be explained by the same or similar subsets of variables to be selected from a large set of candidate variables. For example, we can take a series of cat pictures as the response, and then for a given set of the basis functions, we want to identify the common basis functions for these cat pictures. This kind of problem has also been studied by Lounici, Pontil, Tsybakov, and Van De Geer (2009) under the name of multitask learning, and by Obozinski, Wainwright and Jordan (2011) under the name of support union recovery, where the word "support" means the subset of variables selected for a response vector, and "support union" means the union of subsets of variables selected for all the response vectors.

To deal with the support union recovery problem, Obozinski et al. (2011) set the group structure for each variable across multiple response vectors, and the group LASSO approach was adopted. As mentioned before, we can easily use the Bayesian group selection approach to identify the union of the support sets by setting the indicator to denote whether $\mathtt{X}_i$ is active or not. In addition to the group sparsity, Chen et al. (2014) consider the sparse group sparsity in this support union recovery problem. That is, the selected variable might not be active for all response vectors. In other words, we are interested in finding the best union of support sets, but we also assume that the active variable might be inactive for some response vectors. Thus, following Chen et al. (2016), two nested sets of binary indicator variables are used. The first set of indicators, $\delta = (\delta_1, ..., \delta_p)'$, is associated with variables, $\mathtt{X}_1, ..., \mathtt{X}_p$, respectively, and δ_j is defined to indicate if the variable X_j is active for any of the response vectors. In the second indicator set, each indicator is associated with a variable and a response vector, indicating whether this variable is active for explaining the particular response vector. Thus, for each variable $\mathtt{X}_j$, we define the indicator vector $\eta^{(j)} = (\eta_{j,\,1}, ..., \eta_{j,\,M})$, and if $\eta_{j,\,m} = 1$, the variable $\mathtt{X}_j$ is active for the mth response, $\mathtt{Y}_m$, and $\eta_{j,\,m} = 0$ otherwise.

Similar to the groupwise Gibbs sampler (Chen et al, (2016)), the prior distribution of δ_j is the Bernoulli distribution with $P(\delta_j = 0) = \theta_j$, i.e., $Ber(1 - \theta_j)$. For the prior for the second set of indicators, the prior of $\eta_{j,\,m}$ is chosen as a mixture distribution dependent on the indicator in the first set δ_j and is represented as

$$\eta_{\,j,m} \,|\, \delta_j \sim (1-\delta_j)\gamma_0 + \delta_j Ber\left(1-\rho_{\,j,m}\right), \tag{7}$$

where $P(\eta_{j,\,m} = 0) = \rho_{j,\,m}$. Based on Equation (7), if the jth variable, X_j, is not selected in S, that is, $\delta_j = 0$, then $\eta_{j,\,m} = 0$ for all $m = 1, ..., M$; however, when $\delta_j = 1$, $\eta_{j,\,m}$ still could be 0 or 1 due to the Bernoulli prior distribution. Then, for the coefficient, $\beta_{j,\,m}$, given the indicators δ_j and $\eta_{j,\,m}$, the prior distribution of $\beta_{j,\,m}$ can be defined as

$$\beta_{\,j,m} \,|\, \delta_j, \eta_{\,j,m} \sim \left(1-\delta_j\eta_{\,j,m}\right)\gamma_0 + \delta_j\eta_{\,j,m} N\left(0, \tau^2_{\,j,m}\right). \tag{8}$$

That is, the prior of $\beta_{j,\,m}$ is $N\left(0, \tau^2_{j,m}\right)$ only when $\delta_j = 1$ and $\eta_{j,\,m} = 1$. Then, the groupwise Gibbs sampling can be modified to generate the posterior samples of these indicators for the final Bayesian inference. In Chen et al. (2014), the proposed Bayesian method is used for image data analysis, that is, to identify the commonly used basis functions for a given series of images.

5.2.2 | Structure selection in vector autoregression models

Chu, Lo Huang, Huang, and Chen (2019) proposed a Bayesian-type method for estimating unknown coefficients in the vector autoregressive (VAR) models. Since the total number of coefficients would be quite large, especially when we include more time series, to save the computational cost for coefficient estimation, different coefficient matrix structures are taken into account. Consider the following VAR model with order p,

$$\mathtt{Y}_t = \mathtt{Y}_{t-1}B_1 + \mathtt{Y}_{t-2}B_2 + \cdots + \mathtt{Y}_{t-p}B_p + \varepsilon_t, \quad t = 1, ..., T,$$

where $\mathtt{Y}_t = \left(y_{t,1}, ..., y_{t,M}\right)$ is a $1 \times M$ vector, B_l is an $M \times M$ matrix for $l = 1, 2, ..., p$, and $\{\varepsilon_t\}_{t=1}^{T}$ is a sequence of serially uncorrelated random vectors coming from a multivariate normal distribution with zero mean and covariance matrix Σ.

In many financial data sets, the data generally have some natural structures, and the information of the natural structures would be helpful for estimating the coefficient matrix. Thus, rather than estimating the coefficient matrix either all at once or in a column-by-column manner, Chu et al. (2019) divide these columns into nonoverlapping segments and then attempt to estimate the B_l values segment by segment. Under the segment grouping, two indicators $\gamma_{l,\,m}$ and $\eta_{l,\,m,\,k}$ are used to indicate which coefficients are active for the kth segment in the mth row for B_l with respect to their own lags and others' lags. Similar to Chen et al. (2016), the prior of elements of the coefficient matrix B_l are dependent on γ_l and η_l as follows:

$$B_{l,m,m} \mid \gamma_{l,m} \sim (1-\gamma_{l,m})\delta_0 + \gamma_{l,m} N(0,\tau^2),$$
$$B_{l,m,\tilde{s}_k} \mid \eta_{l,m,k} \sim (1-\eta_{l,m,k})\delta_0 + \eta_{l,m,k} N_{|\tilde{s}_k|}(\mathbf{0},\tau^2 \mathtt{I}),$$

where $\tilde{s}_k$ is the index set of the kth segment except m. They assume the prior of the covariance matrix Σ is from the inverse Wishart distribution with the degree of freedom, λ, and scale matrix, V. To obtain the posterior samples of $\gamma_{l,\,m}$ and $\eta_{l,\,m,\,k}$, a Gibbs sampler procedure is proposed, and basically, this MCMC algorithm is similar to the groupwise Gibbs sampler in Chen et al. (2016). Chu et al. (2019) conducted an empirical study to present structural changes in the stock prices of 18 companies under different economic conditions.

6 | CONCLUSION

Structure selection plays an important role in current statistical model building. In this paper, we focus on group structure, and two types of Bayesian group selection approaches are illustrated. One is the indicator-type method, and another one is generated from the LASSO approach. In addition to the group structure, we also take the sparse group selection into account. That is, we not only select the active groups but also identify the important variables within the selected groups. To illustrate the performance of the Bayesian approaches, we conduct two simulation examples. The results support that the BSGS approaches have lower FPR values and higher TCR values compared to the LASSO-type methods.

In practice, the dependencies among the variables are crucial for variable (structure) selection problems. In our simulations, we did generate correlated variables; for example, in the second scenario, the pairwise correlation among variables is 0.5. However, it seems that the Bayesian approaches can still identify the proper model. In Farcomeni (2010), he considered a strong collinearity situation in a design matrix, but the Bayesian constrained selection method still identified the true model. In addition, in Farcomeni (2010) and Chen et al. (2016), they only required that the limiting correlation matrix is positive definite for their theoretical results, and it is not related to the correlation structures in the design matrix, $\boldsymbol{X}$. Finally, Bayesian approaches should have better numerical stability even when the variables are nearly collinear. This is because the inverse of the covariance matrix for the conditional posterior distribution of β is not only dependent on $\mathtt{X}'\mathtt{X}$ but also adds with a term due to the prior structure. Thus, the posterior mean of β may be regarded as a ridge-type regression estimation. However, more studies related to the effect of the dependence structures among variables should be necessary.

In this paper, we only consider the disjoint group structure. However, in practice, the groups may be overlapping. For example, Zhang, Morris, Zhang, Orlowski, and Baladandayuthapani (2014) studied Bayesian joint selection of genes and pathways with applications in multiple myeloma genomics. In their works, a given pathway is treated as a group of interacting genes, and their goal is to select both pathways and genes that are similar to the sparse group selection; however, the groups may be overlapping. To conduct the selection among overlapping groups and the variables within groups, they proposed an overlap hierarchical structured variable selection (overlap-HSVS) method, and basically, this overlap-HSVS can be treated as a type of the Bayesian indicator selection approach. Due to the ever-increasing complicated data structures, different structure types may occur; therefore, determining how to design the corresponding Bayesian structure selection approaches would be an interesting research problem.

CONFLICT OF INTEREST

The authors have declared no conflicts of interest for this article.

AUTHOR CONTRIBUTIONS

Wei-Ting Lai: Methodology; writing-original draft. **Ray-Bing Chen:** Methodology; writing-original draft; writing-review and editing.

ORCID

Ray-Bing Chen https://orcid.org/0000-0001-7226-509X

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of this article.

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